

Final Endeavor for the Generalized t -SNE Problem

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1 Progress So Far and Setup

OK this is it. We have reached this step and the end is within our grasp. The stake is high and I can literally feel the adrenaline coursing my vein. So far we have done two things. First, we have finished the proof of convergence for the generalized output kernel. The main theorems in [1] will hold if

Condition 1 (Output Kernel Conditions). *The output kernel $g(y, y') = k(\|y - y'\|)$ depends only on the distance between y and y' and that:*

1. $k : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ is decreasing and bounded;
2. g satisfies

$$\iint_{(0, \infty)^2} g(y, y') dy dy' < \infty; \quad (1.1)$$

3. k has bounded derivative;
4. $k'(0) = 0$.

Moreover, the lemmas that deal with the output does not interfere with those related to the input kernel. The initial exploration we did around the output kernel not only provides us a good opportunity to understand the problem and the literature, but also serves as an integral part of the final paper. If we are able to tackle the lemmas related to the input kernel we will be able to join the results seamlessly.

On the other hand, we have established the form of the generalized input kernel, set a basis for us to work on, and prove the existence of $\sigma_{\rho, \mu}^*(x)$, the unique solution to $F_{\rho, \mu}(x, \sigma) = 0$. For this to happen, the generalized input kernel should satisfy the following conditions:

Condition 2 (Initial Input Kernel Conditions). *Let $w : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ be some fixed function. The input kernel has the form of $\exp(-w(\sigma_i \|x_j - x_i\|^\theta))$. Here, σ_i is used to control entropy (thus perplexity) and $\theta > 0$ is a fixed constant. (We can replace σ_i with $\psi : \mathbb{R}^d \rightarrow \mathbb{R}^+$ in the continuous version of the kernel.) w satisfies:*

1. $w'(t) > 0$ for all $t \geq 0$ with $\lim_{t \rightarrow \infty} w(t) = \infty$;
2. $w'(t) + tw''(t) \geq 0$ for all $t \geq 0$;
3. The following two integrals converge:

$$\int_0^\infty t^{d-1} \exp(-w(t^\theta)) dt < \infty, \quad \int_0^\infty t^{d-1} w(t^\theta) \exp(-w(t^\theta)) dt < \infty. \quad (1.2)$$

Since the probability masses are all normalized, we can assume without loss of generality that $w(0) = 0$. Note that condition 2 is only enough to prove the existence and uniqueness of σ^* . We may add more assumptions later on if necessary. Before we march towards the final battle, let us remind ourselves some important definitions and notations, as well as some of the lemmas from [1] that we can borrow. Recall that X or the corresponding probability measure μ is called sub-Gaussian if $\mathbb{P}(X > t) \leq 2 \exp(-Ct^2)$ for some positive constant C .

Lemma 1. *Let ν be sub-Gaussian with second moment equal to 1 and suppose $X \in \mathbb{R}^d$ has i.i.d. entries drawn from ν . Then there exists constant C, c_1 such that*

$$\mathbb{P}(\|X\| \geq \sqrt{d} + t) \leq Ce^{-c_1 t^2}. \quad (1.3)$$

Lemma 2. *Let μ be sub-Gaussian and c_1 be the constant from lemma 1. Let $X_1, \dots, X_n \in \mathbb{R}^d$ have i.i.d. entries drawn from μ . Then there exists a constant C such that*

$$\mathbb{P}\left(\max_{i=1, \dots, n} \|X_i\| \geq \sqrt{\frac{\log n}{c_1}} + \sqrt{d} + t\right) \leq Ce^{-c_1 t^2}. \quad (1.4)$$

Lemma 3. *Let $\delta > 0$ be constant, $K_n = [\delta, \sqrt{n}]$ and $h : \mathbb{R}^d \times \mathbb{R}^d \times \mathbb{R} \rightarrow \mathbb{R}$. Suppose the following hold:*

1. μ is sub-Gaussian;
2. $h(x, x', \sigma)$ and $\nabla_x h(x, x', \sigma)$ are bounded for all x, x' and all $\delta \in K_n$;
3. $h(x, x', \sigma) \rightarrow 0$ as $\|x - x'\|/\sigma \rightarrow \infty$.

Let $X_1, \dots, X_n$ be i.i.d. random variables drawn from μ . Let $\mu_n = (1/n) \sum_i \delta_{X_i}$. Then for any $\epsilon > 0$ with probability at least $1 - e^{-c\epsilon^2 n}$ we have

$$\sup_{\sigma \in K_n, x} \left| \int h(x, x', \sigma) d\mu_n(x) - \int h(x, x', \sigma) d\mu(x') \right| \leq \epsilon + O\left(\sqrt{\frac{\log n}{n}}\right). \quad (1.5)$$

Finally, let us also recall that

$$\begin{aligned} F_{\rho, \mu}(x, \sigma) &= \int \frac{\exp(-w(\sigma \|x - x'\|^\theta))}{\int \exp(-w(\sigma \|x - x'\|^\theta)) d\mu(x')} \log \left(\frac{\exp(-w(\sigma \|x - x'\|^\theta))}{\int \exp(-w(\sigma \|x - x'\|^\theta)) d\mu(x')} \right) d\mu(x') + \log \rho \\ &= - \int \frac{w(\sigma \|x - x'\|^\theta) \exp(-w(\sigma \|x - x'\|^\theta))}{\int \exp(-w(\sigma \|x - x'\|^\theta)) d\mu(x')} d\mu(x') \\ &\quad - \log \left(\int \exp(-w(\sigma \|x - x'\|^\theta)) d\mu(x') \right) + \log \rho. \end{aligned} \quad (1.6)$$

The two terms in the formula of F are the pillars of this manuscript. Their convergence will lead us to prove the convergence of F , and most importantly, the uniform convergence of $\sigma(x)$.

2 The Two Most Important Lemmas

OK I am nervous as fuck but let us do this.

3 The Rest of the Proof

4 Wrap Up and Conclusion

References

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